CS726: Assignment-1

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1 Introduction

We implememt the message passing algorithms for computing marginal probabilities and the **Maximum A Posteriori (MAP)** assignment in an **undirected graphical model (UGM)**. The process includes processes like graph triangulation, junction tree construction, etc. **References like chatgpt and online sources were used for this assignment and have been cited in comments** .

2 Graph Triangulation

- Triangulation converts the given undirected graph into a chordal graph.
- Ensures that the graph can be represented as a junction tree.
- Minimum Degree Heuristic is used for triangulation, which selects the node with the smallest degree iteratively and eliminates it.

3 Junction Tree Construction

- After triangulation, the junction tree is constructed by grouping the maximal cliques of the triangulated graph.
- The tree is structured to maintain the running intersection property, ensuring consistency in the belief propagation process.
- Prim's algorithm is used to generate the Maximum Spanning Tree (MST) from the cliques.
- Negative of separator set sizes are used as edge weights in Prim's algorithm to determine the MST.

4 Z value computation

The belief at clique C_r is given by:

$$\tilde{\psi}_{C_r} = \psi_{C_r} \prod_{S \in \text{Neigh}(C_r)} m_{S \to C_r}$$

where:

- ψ_{C_r} is the initial potential of the clique.
- $m_{S \to C_r}$ are the messages received from neighboring separators.

Sum Over All Variables in C_r

The partition function Z is then:

$$Z = \sum_{x_{C_r}} \tilde{\psi}_{C_r}(x_{C_r})$$

where the sum marginalizes over all variables in C_r.

5 Message Passing Algorithms

5.1 Sum-Product Algorithm for Marginal Probabilities

- Computes the marginal probability of a variable by propagating messages through the junction tree.
- Messages are computed as:

$$m_{C_j \to C_i}(X) = \sum_{Y} \phi(X, Y) m_{C_k \to C_j}(Y) \tag{1}$$

- X and Y are variables in cliques C_i and C_j, and $\phi(X, Y)$ represents the potential function.

5.2 Max-Product Algorithm for MAP Assignment

- Finds the most probable assignment of values to variables.
- Follows the same structure as the sum-product algorithm but replaces summation with maximization:

$$m_{C_j \to C_i}(X) = \max_{Y} \phi(X, Y) m_{C_k \to C_j}(Y) \tag{2}$$